



Society of Petroleum Engineers

SPE-229403-MS

Building Resilient Safety Culture: PDO Integrated HSE Coaching and Competency Assurance Model Transforming Contractors and Organizational Leadership

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Petroleum Development Oman (PDO) has implemented a new Health, Safety, and Environment (HSE) coaching program focused on transforming safety behaviors and culture among Major Contractors, Local Community Contractors (LCCs), Small and Medium Enterprises (SMEs), and PDO internal leadership workforce occupying Safety Critical Positions (SCP). This initiative supports PDO's broader Goal Zero ambitions, aiming for zero harm and incidents across its operations.

PDO's HSE coaching program engages contractors and internal staff through structured group and one-on-one coaching sessions designed to build individual and organizational capability, foster safety ownership, and align external safety cultures with PDO's expectations. Simultaneously, PDO applies the Lead/Prepare/Apply (LPA) risk management framework to internal leadership. The LPA system ensures that leaders are equipped with requisite HSE knowledge, skills, and behaviors to lead by example and coach their teams effectively. This dual strategy harmonizes capability development across the operator and contractor ecosystem, driving safer work practices and enhanced risk management.

Introduction

Petroleum Development Oman (PDO) has implemented a new Health, Safety, and Environment (HSE) coaching program focused on transforming safety behaviors and culture among Major Contractors, Local Community Contractors (LCCs), Small and Medium Enterprises (SMEs), and PDO internal leadership workforce occupying Safety Critical Positions (SCP). This initiative supports PDO's broader Goal Zero ambitions, aiming for zero harm and incidents across its operations.

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Literature Context and Theoretical Underpinning

The theoretical foundation for this integrated model draws on multiple established frameworks:

- Safety Culture Theories: [Zohar \(1980\)](#) and [Guldenmund \(2000\)](#) emphasize the importance of shared values, norms, and practices in shaping workplace safety
- Behavioral Safety and Coaching Literature: [Geller \(2005\)](#) and [Nielsen \(2014\)](#) highlight how coaching and behavioral interventions improve safety climate and reduce injury rates
- Competency Frameworks: The [Australian Institute of Health & Safety \(2020\)](#) stresses leadership and human factors as critical elements in safety management
- Organizational Resilience: [Reason's \(1997\)](#) model supports a shift from compliance toward capability, highlighting proactive risk management and leadership engagement as key to preventing incidents.

Building on these theoretical foundations, PDO operationalized its vision through a company-wide coaching initiative that merges behavioral theory, adult learning principles, and competency assurance. Empirical evidence supports coaching as a transformative tool for organizational culture change. For example, [Nielsen \(2014\)](#) found that coaching safety professionals enhanced their engagement, improved the safety climate, and lowered injury rates.

Coaching is widely regarded as a high-impact developmental intervention, particularly in high-reliability, complex industries such as oil and gas ([Grant, 2014](#); [Jones et al., 2016](#)). At PDO, coaching and mentoring have been pivotal in the Journey of Transformation (JoT), delivering measurable improvements in leadership behavior, safety performance, and operational discipline.

As part of this strategy, PDO developed and scaled a large internal coaching capacity through the International Coaching Federation (ICF) and the Association of Coaching (AoC) aligned training programs. Nearly 40 coaches were certified and more than 200 leaders trained in coaching skills by 2024 ([Talogy, 2024](#)). This internal capability was complemented by external partnerships and recognized through regional awards, including the CIPD Middle East "Best Coaching & Mentoring Initiative" award ([Talogy, 2024](#)).

PDO's Coaching & Mentoring Framework

In 2024, PDO reaffirmed its solid commitment to HSE excellence by revising its HSE Policy under the direction of the Managing Director to explicitly incorporate HSE coaching and mentoring as a core strategic pillar. As part of PDO's Journey of Transformation (JoT), this update signifies a pivotal shift from a compliance driven mindset to a culture of proactive engagement, leadership development, and continuous learning. By embedding coaching and mentoring across all levels, from frontline workers to executive leadership, PDO has institutionalized a learner centric approach that empowers individuals to take ownership of safety. The policy now underscores collective participation in coaching as a key driver of sustainable HSE improvement, fostering psychological safety, strengthening risk awareness, and enabling peer-to-peer influence. This top-down endorsement sends a powerful message across PDO and its contractor community that safety is not just a rule to follow, but a value to be lived through guided development, active listening, and shared accountability.

This strategic commitment laid out in the revised HSE Policy has been operationalized through the development of a structured and organization-wide Coaching & Mentoring Framework. Recognizing that policy alone is insufficient without consistent practice, PDO translated its top-down endorsement into a formalized system designed to embed coaching and mentoring across all layers of the workforce.

To ensure systematic application and sustainable impact, PDO formalized a Coaching & Mentoring Framework. This framework provides guidelines for coaching implementation across all levels, supporting new graduates, experienced professionals, and senior leadership. It employs diverse coaching techniques

(developmental, situational, and performance-based) and mentoring models that integrate both technical and behavioral development goals (Passmore & Fillery-Travis, 2011).

Aligned with adult learning theory and global HSE leadership standards, the framework seeks to:

- Enhance task-specific and leadership competence
- Identify and close gaps in safety processes, capability, and learning
- Promote reflective practice and proactive risk mitigation
- Build leadership identity and psychological safety

HSE Coaches play a central role within this structure, identifying risks stemming from poor implementation of controls, ineffective communication, or declining team performance. They work closely with site-based Safety Critical Positions and contractor staff to promote situational awareness, learning from normal work, and intervention before risk becomes harm.

This integrated coaching model complements PDO's capability assurance system, ensuring that technical knowledge translates into behavior, and that leaders at every level are equipped to influence safe operations effectively (Dekker, 2014; Reason, 1997).

Overview of PDO's HSE Coaching Program

Program Launch and Intent

PDO launched its dedicated HSE coaching initiative in 2023 to transform safety behaviors among its contractors and PDO internal leadership, reinforcing the Goal Zero agenda through behavioral capability development. The program was designed not just to provide safety instruction, but to facilitate behavior change by fostering ownership, reflection, and learning.

Dual Coaching Strategy

The program operates through two complementary streams:

Contractor Coaching (Major Contractors, LCCs, SMEs). This external stream targets frontline contractor personnel. Coaches deliver both group-based and one-on-one sessions, focused on improving risk perception, safety engagement, and compliance behaviors. Tailored to the contractors' varying maturity levels, coaching sessions emphasize practical HSE application, reflective dialogue, and continuous improvement.

Internal Leadership Coaching (LPA Framework). Internally, PDO adopted the Lead/Prepare/Apply (LPA) framework to enhance leadership capability across departments. Safety Critical Positions are identified, and coaching is tailored to assess, support, and advance competencies required to lead by example. Sessions draw on ICF/AoC-certified methods and include follow-ups to track behavioral change and field application.

Together, these streams ensure that both operators and contractors develop aligned safety mindsets and a shared language of risk management.

Coaching Content and Capability Framework

To support structured development, PDO created the HSE Capability Model, comprising 30 modules grouped into six core capability domains. These are designed to address both technical competence and behavioral leadership needs, mapped to PDO's "Know, Say, Do" model:

- **Know:** Foundational knowledge of safety principles and systems

- **Say:** Communication skills to articulate safety insights and influence teams
- **Do:** Consistent application of safe practices and proactive risk interventions

Six Capability Domains + Two Leadership Domains

The coaching content derives from the PDO HSE Capability Model as shown in Figure 1, which breaks down HSE capability into 30 modules spanning technical knowledge, interpersonal skills, risk psychology, leadership, and behavioral competencies (PDO HSE Coaching Department, 2024).

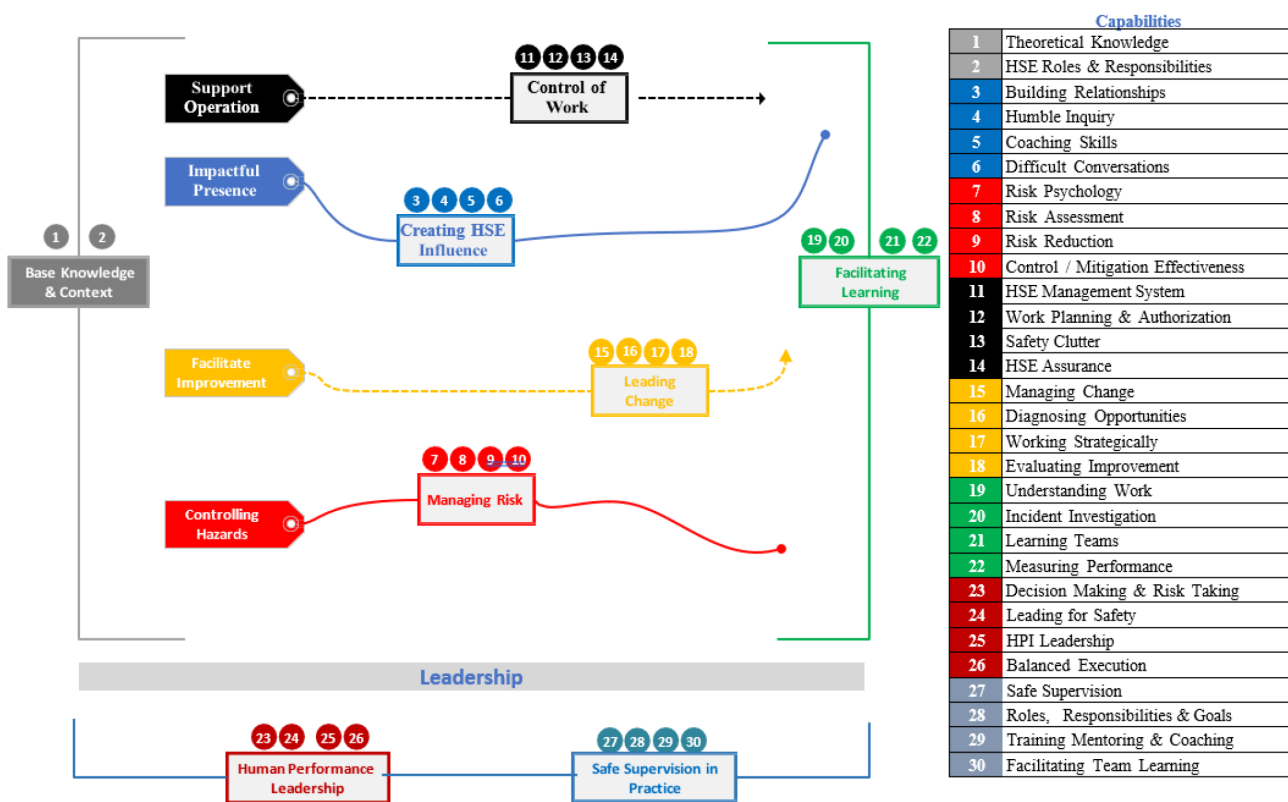


Figure 1—PDO HSE Capability Model

Base Knowledge and Context

Focus: Foundational theory and role clarity

- Targets: Primarily HSE professionals; context also relevant for operations personnel
- Components include understanding of HSE theory, legislation, organizational policies, and clarity on HSE-related roles and responsibilities

Control of Work

Focus: Safe planning, authorization, and execution of tasks

- Targets: HSE professionals and business leaders in the safety critical positions
- Emphasizes effective integration of HSE considerations in work planning and execution, supporting the implementation of the HSE Management System

Creating Influence

Focus: Behavioral influence and soft skills

- Targets: HSE professionals
- Recognizes that HSE personnel often rely on influence (rather than authority) to promote safe decisions. Coaching focuses on interpersonal dynamics, trust-building, and persuasive communication.

Leading Change

Focus: Improvement and change management

- Targets: Business leaders and HSE professionals
- Encourages a proactive mindset for recognizing and acting upon opportunities for HSE improvement, with structured follow-through and impact evaluation.

Managing Risk

Focus: Risk-based thinking and controls

- Targets: HSE professionals and operational leaders
- Promotes deep understanding of hazard identification, risk assessment, and barrier effectiveness, driving consistent decision-making based on risk intelligence.

Facilitating Learning

Focus: Learning from work (normal and abnormal)

- Targets: HSE professionals and operational teams
- Advocates for a learning-focused safety culture by investigating incidents, understanding everyday operations, and facilitating knowledge sharing through learning teams and reflective practices.
- Leadership-Specific Capability Domains
PDO also emphasizes leadership development through two applied capability domains focused on supervisory and strategic safety leadership.
 - A. Human Performance Leadership
Modules:
 - Decision Making & Risk Taking
 - Leading for Safety
 - HPI (Human Performance Improvement) Leadership
 - Balanced Execution

These modules equip leaders with the mindset and tools to model safe behavior, balance competing priorities, and foster psychologically safe environments where employees feel empowered to speak up.

B. Safe Supervision in Practice

Modules:

- Safe Supervision
- Roles, Responsibilities, and Goals
- Training, Mentoring, and Coaching

- Facilitating Team Learning

This area focuses on frontline leadership, developing the ability to coach teams, align expectations, and reinforce safe operational routines. These modules form the content backbone for coaching sessions and are designed to be delivered through interactive group workshops, one-on-one coaching, and field-based application. Each module integrates behavioral science, adult learning theory, and practical HSE application.

Sessions typically involve:

- Group Workshops: Interactive modules encouraging discussion, scenario analysis, and sharing of experiences
- One-on-One Coaching: Personalized sessions aimed at reinforcing learning, addressing specific challenges, and developing actionable improvement plans

PDO employs 23 qualified HSE coaches distributed across operational clusters, enabling localized delivery and contextual relevance.

Scale and Reach

From 2024 to June 2025, the program delivered over 3,000 coaching sessions with approximately 40,000 coachees engaged. Around 10,000 coachees were PDO employees, while the majority were from PDO contractors. This extensive reach ensures wide-scale impact and fosters alignment of safety culture across the entire operational ecosystem.

Material and Methods

This study employs a mixed-methods approach, combining quantitative analysis of coaching session data with qualitative evaluation of program design, delivery processes, and observed behavioral changes.

Data Sources

- Coaching session records
- Competency assurance metrics
- Participant feedback surveys

The methodology was designed to assess the reach, engagement, and effectiveness of the HSE coaching program for Major Contractors, LCCs, and SMEs, while simultaneously evaluating the internal leadership coaching through the LPA risk management framework.

The framework for analysis draws on:

- Safety climate theory
- Capability development models
- Organizational learning concepts

This integrated perspective enables the assessment of how coaching and structured competency development contribute to a resilient and proactive safety culture.

Organizational Coaching Transformation and Leadership Development

In parallel with its HSE coaching strategy, PDO launched a broader transformation to embed coaching into its leadership culture. This transformation journey was designed to enhance internal leadership effectiveness and create a psychologically safe, feedback-driven environment across all management levels.

Recognizing the need for a resilient and adaptable leadership pipeline, PDO initiated a strategic partnership with a Consultancy to design and deliver a scalable internal coaching infrastructure. The initiative was guided by two core objectives:

1. To develop a pool of certified internal coaches capable of delivering context-sensitive coaching aligned to PDO's operational realities
2. To implement a customized "Leader as Coach" program that equips PDO supervisors and managers with essential coaching capabilities

Step 1: Building Accredited Internal Coaches

PDO launched two accredited coaching tracks: Foundation Executive Coach and Executive Coach, both endorsed by the International Association for Coaching. A selected cohort of 24 participants completed rigorous, year-long training that included:

- Over 250 hours of internal coaching practice
- 80 hours of self-directed learning
- 40 hours of group-based instruction and live assessment

This hands-on approach enabled coaches to apply theoretical concepts directly within operational environments, supported by structured feedback and reflection mechanisms. The curriculum was tailored to PDO's unique culture and challenges, including the complexities of the Omani energy sector.

Step 2: Rolling Out the 'Leader as Coach' Program

Following the development of the internal coach cadre, PDO and Consultant launched the Leader as Coach program. This three-day immersive course, certified by the Association for Coaching, was designed to cultivate essential coaching behaviors among line managers and supervisors. Key learning components included:

- Advanced questioning techniques to foster independent thinking and problem-solving
- Principles of psychological safety to encourage open dialogue and team learning
- Simulated coaching sessions with real-time feedback from expert facilitators
- Peer coaching models to maintain coaching behaviors beyond formal settings

To ensure long-term sustainability, a Train-the-Trainer model was implemented, enabling five internal coaches to facilitate the program with high quality and fidelity. This internal capability supports wide-scale delivery of leadership coaching across PDO, strengthening ownership and adaptability.

Strategic Alignment with HSE Capability Goals

This leadership coaching initiative is a direct complement to PDO's HSE coaching and mentoring framework. Both programs are unified by the "Know, Say, Do" model, which serves as a foundation for translating learning into real-world behaviors that enhance HSE and performance.

By embedding coaching principles at both field and leadership levels, PDO ensures that technical knowledge, interpersonal influence, and psychological insight are developed together. These capabilities are essential for navigating the complexities of high-risk operations, aligning closely with literature on high-reliability organizations (Weick and Sutcliffe, 2015).

Moreover, the internal coaching pool enhances PDO's capacity to support ICV through Omanisation and local capability growth. It reduces dependency on external consultants while creating a sustainable structure for ongoing leadership development, cultural alignment, and safety performance improvement.

Coaching Approach

Coaches facilitate learning through the following techniques:

- Asking Open-Ended Questions: To stimulate reflection, critical thinking, and problem-solving, encouraging coachees to analyze their own situations
- Maintaining Active Listening and Silence: Creating space for coachees to take ownership of discussions and solutions, reinforcing their confidence
- Collaborative Agreement on Coaching Issues: Ensuring that the coachee and coach jointly identify and agree upon coaching topics before exploring problems in-depth
- Avoiding Direct Advice-Giving: Encouraging coachees to develop their own solutions and empowering them rather than imposing ready-made answers
- Using Follow-Up Observations and Opportunistic Coaching: Reinforcing learning through real-time coaching in the workplace, helping coachees apply lessons directly in their operational context

This approach integrates behavioral safety principles, adult learning theories, and safety culture models to ensure lasting behavior change (Geller, 2005; Nielsen, 2014; Provan et al., 2017). Coaches serve as facilitators rather than instructors, promoting active engagement and ownership among participants.

Participant Selection and Target Groups

The study focuses on two main populations:

- Contractors (Major, LCCs, SMEs): Selected from PDO's contractor database based on HSE criticality, operational maturity, geography, and past performance indicators
- PDO Internal Leadership: This group includes HSE professionals, operational supervisors, and business leaders who occupy Safety Critical Positions. Participants were identified based on their roles in safety leadership and their inclusion in the competency assurance program

Data Collection Instruments

Multiple data collection instruments were used to capture the diverse dimensions of the program's effectiveness:

- Coaching Session Records: Documentation of group and one-on-one sessions including attendance, coaching topics covered, and coaching prompts used during follow-up sessions
- Competency Assessments: Results from the LPA risk management framework include post-coaching evaluations, leader self-assessments, and assessor ratings
- Safety Performance Metrics: Operational safety indicators such as incident rates, near-misses, and safety climate survey results collected after the program implementation
- Participant Feedback Surveys: Structured questionnaires and open-ended feedback from coachees and coaches on program relevance, effectiveness, and areas for improvement

Quantitative Data Analysis

Data collected from coaching sessions and competency assessments were compiled and analyzed using descriptive and inferential statistics to identify participation trends, competency improvements, and safety performance impacts.

- **Participation Rates:** Attendance was tracked by contractor type, cluster location, and participant role to assess the reach and engagement level of the program
- **Competency Improvement:** Post-coaching feedback scores were analyzed to assess the significance of improvements in HSE knowledge, leadership behaviors, and coaching effectiveness among PDO internal leaders
- **Safety Performance Correlation:** Relationships between coaching engagement levels and safety incident rates were examined. This sought to identify whether higher coaching participation was associated with reductions in incidents and improved safety metrics

Qualitative Data Analysis

Open-ended participant feedback collected from surveys was subjected to thematic analysis to extract recurring themes related to program strengths, challenges, and perceived outcomes. This qualitative analysis provided rich contextual understanding, informing program refinement and highlighting successful elements such as:

- Enhanced safety communication
- Increased leadership engagement
- Greater ownership of safety responsibilities

Data Presentation

The study's findings are presented through a combination of quantitative and qualitative formats for clarity and depth:

- **Summary Tables:** Displaying the number of coaching sessions delivered by population group and the corresponding number of coachees
- **Graphs and Charts:** Visualizing coaching feedback score improvements post-coaching across the program timeline
- **Qualitative Quotes:** Illustrating participant experiences and reflections to provide a human perspective on the coaching program's impact

Summary Table: Coaching Sessions and Participant Breakdown

The table demonstrates the extensive participation of PDO's contractor and internal leadership cohorts in the coaching program. Major contractors represent the largest portion of both sessions and coachees, reflecting their significant operational presence. The inclusion of LCCs and SMEs, although smaller in absolute numbers, shows targeted efforts to engage a diverse contractor base. PDO's Safety Critical Positions account for nearly half of the sessions and a quarter of the coachees, underscoring the focus on leadership development.

This wide distribution highlights the program's comprehensive reach and balanced engagement strategy, which integrates external contractors and internal leaders to drive a consistent safety culture.

Presentation of Data and Results

This section presents the quantitative and qualitative outcomes derived from PDO integrated HSE coaching program. The data illustrate participation metrics, competency improvements, behavioral and cultural shifts, and early indicators of enhanced safety performance across major Contractors, LCCs, SMEs, and internal PDO leadership through the LPA risk management framework.

Participation and Reach of Coaching Sessions

Between January 2024 and June 2025, the program delivered over 3,000 coaching sessions, engaging approximately 40,000 coachees across multiple cohorts. [Table 1](#) (in Part 3) detailed the distribution of sessions and participants. Month wise breakdown of coaching sessions are represented in [Figure 2](#).

- Major Contractors: Delivered 1,500+ sessions, engaging over 30,000 individuals, accounting for the majority (75%) of total participants. These contractors represent large, established organizations with more mature HSE systems but benefitted from the coaching's focus on behavior and culture.
- Local Community Contractors and SMEs: Despite representing a smaller share of sessions (30+), these sessions engaged over 1,300 coachees. Given their diverse operational maturity and resource constraints, targeted coaching is critical to elevate their HSE capabilities and align their practices with PDO standards.
- PDO Safety Critical Positions: With 1,400+ sessions and 10,000 coachees, this internal cohort includes operational leaders and HSE professionals who are key to driving safety initiatives. The coaching here focuses on leadership competencies and embedding a coaching culture.

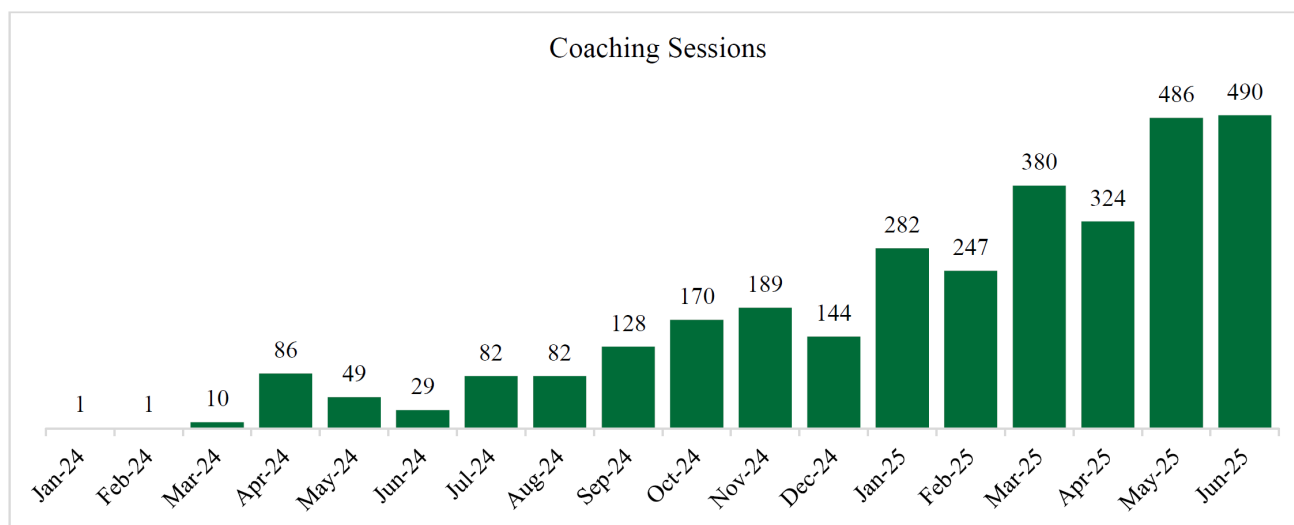


Figure 2—Distribution of Coaching Sessions / Month wise (PDO HSE Coaching Report, 2025)

Table 1—Distribution of Coaching Sessions and Participants (PDO HSE Coaching Report, 2025)

Population Group	Number of Group and One-to-One Coaching Sessions	% of Total Coaching Sessions	Number of Coachees	% of Total Coachees
PDO Major Contractors	1,500+	50%	30,000+	75%
LCCs & SMEs	30+	1%	1,300+	3.25%
PDO Safety Critical Positions	1,400+	46.7%	10,000+	25%
Total	3,000+	100%	40,000+	100%

Localized coaching clusters and flexible scheduling facilitated effective participation, integrating coaching into operational routines. Feedback from coaches emphasized that having accessible, on-site sessions contributed significantly to attendance and engagement.

Competency Improvements via LPA Framework

As part of its HSE competency assurance process, PDO has introduced the Lead/Prepare/Apply (LPA) framework, which focuses on three core HSE capabilities: Lead, Prepare, and Apply. These competencies are critical for individuals in HSE leadership and safety-critical roles, ensuring they can effectively lead HSE efforts, prepare for HSE risks, and apply HSE standards in daily operations. To support this framework, PDO has certified over 150 HSE assessors from its workforce and more than 300 from its contractor base, equipping them to assess and validate these competencies across the organization through dedicated workshops and structured training programs. The LPA risk management framework was used to assess PDO leadership capabilities before and after coaching interventions.

- Key Competency Dimensions Assessed:
 - HSE Technical Knowledge
 - Leadership Behaviors
 - Coaching Effectiveness
 - Risk Management Skills
- Leadership Feedback:
 - Post-coaching self-assessments reflected increased confidence in leading safety conversations, identifying risks proactively, and supporting team members' development. These enhancements align with PDO's goal of fostering safety leadership beyond compliance.

Behavioral Change Indicators from Contractor Coaching

Coaching sessions with contractors focused on practical safety behaviors and communication. Post-coaching feedback revealed notable positive shifts:

- Self-Reported Behavioral Changes:
 - Over 30% of coachees reported increased confidence in:
 - Speaking up about hazards.
 - Applying risk controls effectively.
 - Participating actively in safety discussions and toolbox talks.
- Observed Changes:
 - Supervisors and PDO observers noted increased hazard reporting, more frequent peer-to-peer safety conversations, and greater compliance with safety protocols during site visits following coaching interventions.

These behavioral shifts are crucial precursors to embedding a sustainable safety culture, emphasizing proactive risk management rather than reactive compliance.

Case Study: Major Contractor Safety Culture Transformation

One major contractor with a history of low safety compliance participated in a focused, intensive 12-month coaching program. The results provide a compelling example of coaching's impact:

- Incident Reduction: Recordable incidents decreased by 40% compared to the previous year, demonstrating enhanced risk awareness and control implementation
- Near-Miss Reporting: Near-miss reports increased by 150%, indicating a growing culture of hazard recognition and transparent communication
- Leadership Visibility: Safety observers documented increased leader presence on-site, with frequent coaching conversations noted. This visibility was credited with reinforcing safety priorities and engaging frontline teams
- Management Interviews: Interviews with contractor management highlighted a stronger ownership of safety processes and an increased motivation to sustain improvements beyond the coaching period

This case exemplifies how targeted coaching interventions can shift entrenched safety cultures and lead to tangible performance improvements.

Qualitative Feedback from Participants

Insights from over 2,500 participant responses reveal key themes reflecting the program's value:

- Diverse Roles Engaged:
 - Supervisors: 930 participants
 - HSE Professionals: 515 participants
 - Technicians: 554 participants
 - Team Leaders: 245 participants
 - Operators and Frontline Workers: 336 participants

This broad representation affirms the program's reach across frontline and leadership roles, vital for system-wide culture change.

- Value of Interactive Learning: Participants appreciated the coaching's interactive, discussion-based format compared to traditional classroom lectures. This engagement style facilitated deeper reflection and practical application
- Increased Ownership: Many expressed feeling empowered to take responsibility for safety, moving beyond viewing it as a compliance burden to an active part of their daily work
- Improved Trust and Collaboration: Contractors noted that coaching strengthened trust and communication with PDO representatives, enabling more cooperative safety management
- Leadership Development: Internal leaders highlighted coaching's role in enhancing their influence and ability to engage frontline teams effectively

Sample Participant Quotes

"The coaching sessions helped me see safety from a new perspective. It's not just about rules but how we engage and influence others." LCC/SME Worker

"Having a HSE coach onsite who understands our challenges made it easier to ask questions and improve our daily HSE practices." Major Contractor Supervisor

These quotes illustrate how coaching fostered new mindsets and improved practical safety application.

Qualitative Feedback Deep Dive

Beyond the broad themes previously discussed, deeper qualitative insights from over 2,500 participants provide a rich understanding of the coaching program's strengths and areas for refinement.

- **Interactive Learning Environment:** Participants consistently valued the practical, discussion-based nature of coaching. Unlike traditional lecture-style training, coaching allowed them to voice concerns, share experiences, and engage in problem-solving. This approach enhanced learning retention and encouraged personal ownership of safety behaviors
- **Empowerment and Ownership:** Many coachees noted a shift in mindset, from compliance-driven to proactive engagement with safety. The coaching process helped them see themselves as active contributors to safety culture rather than passive rule followers. This empowerment fostered increased confidence in raising hazards and influencing peers
- **Trust and Communication:** Contractor respondents reported improved trust between their teams and PDO representatives, attributing this to the regular, respectful interactions fostered by coaching. Enhanced trust reduced resistance and increased openness in discussing safety issues
- **Leadership Growth:** Internal leaders described coaching as instrumental in developing skills to motivate and influence frontline workers. Leaders highlighted improved confidence in conducting safety conversations and coaching others effectively

These qualitative insights align with behavioral safety and adult learning theories emphasizing engagement, reflection, and social learning as key to sustained change (Geller, 2005; Nielsen, 2014).

Coach Ratings and Participant Satisfaction

Participant feedback included ratings of coaching effectiveness based on knowledge delivery, explanation clarity, and facilitation skills are represented in Figure 3. From a total of 2,580 responses:

- Excellent: 1,679 (65%)
- Very Good: 706 (27%)
- Good: 174 (7%)
- Fair: 10 (<0.5%)
- Poor: 11 (<0.5%)

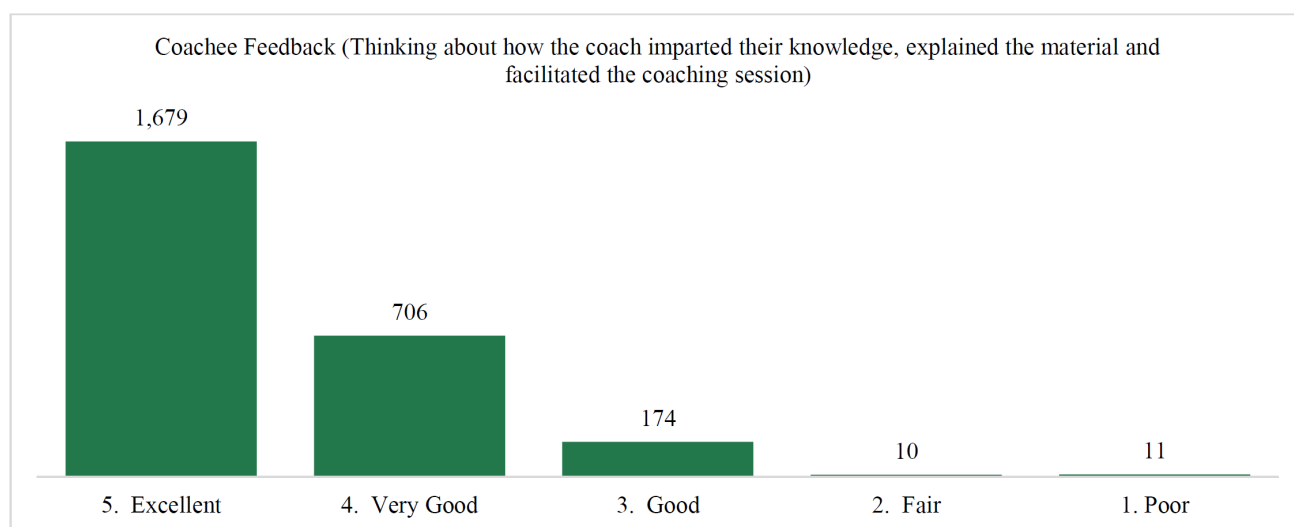


Figure 3—Coachee Feedback on Coaching Delivery and Facilitation Effectiveness

The high percentage of "Excellent" and "Very Good" ratings reflects strong coach performance and participant satisfaction. Minor improvement areas, indicated by "Good" and below, primarily related to coaches' pacing or tailoring content to specific cluster or the operational contexts.

The coaching team used this feedback for continuous professional development, ensuring consistency and relevance across clusters.

Analysis of Lower Ratings and Corrective Actions

While the number of "Poor" (11 or 0.43%) and "Fair" (10 or 0.39%) ratings represents a very small fraction (less than 1%) of total responses, understanding the reasons behind these lower scores remains important. Due to the anonymous nature of the surveys, it is challenging to pinpoint exact causes. However, we believe some lower ratings may stem from language barriers, particularly among contractor personnel who may have limited English proficiency. Additionally, variations in coaching style and session pacing could have influenced participant experience.

To address these issues, PDO is implementing further awareness sessions to better communicate the purpose and benefits of coaching. Efforts are also underway to enhance coaches' cultural competence and language skills to better engage diverse workforce segments. Continuous monitoring and participant feedback mechanisms are used to tailor coaching delivery, and the development of multilingual support materials is being considered to improve accessibility and understanding.

Safety Performance Trends

Early operational data indicate positive correlations between coaching engagement and safety performance improvements. While PDO's formal HSE Coaching Program was launched in early 2024, a pilot phase involving 2–3 coaches began as early as 2021, focusing on selected clusters and high-risk contractor groups. Insights and outcomes from this pilot helped shape the structure and scale of the official coaching initiative.

The trends observed since 2021 reflect the cumulative impact of both the pilot and full-scale rollouts:

- **Significant Fatality Reduction:** Fatalities have decreased by over 70% since 2021, indicating substantial progress in life-critical safety outcomes.
- **SIF Elimination:** Serious Injury, Illness, and Fatality (SIF) incidents were reduced to zero in two of the past three years, highlighting the impact of targeted interventions.
- **LTI Improvement:** Lost Time Injuries (LTIs) have dropped by more than 70% as of mid-2025, demonstrating increased operational safety and workforce protection.
- **HiPo Reduction:** High Potential Incidents (HiPos) have declined by over 85%, reflecting stronger hazard recognition and prevention of near-catastrophic events.
- **Cultural Shift:** Coaching has fostered a culture of proactive safety ownership, enabling workers and leaders to consistently identify and act on risks.
- **Contractor Engagement:** Improvements reflect not only internal leadership changes but also increased confidence and capability among contractor personnel.
- **Coaching as a Strategic Driver:** HSE coaching has emerged as a critical enabler in PDO's transformation toward a zero harm safety culture.

While these trends are promising, longer-term data collection is essential to confirm sustained impact and account for external factors influencing safety outcomes.

Reinforcing Learning through Workplace Coaching

The program emphasized on-the-job coaching rather than isolated classroom sessions. Coaches used:

- Open-Ended Questions: To stimulate reflection and problem-solving without prescribing solutions
- Active Listening and Silence: To encourage coachee ownership and deeper insight
- Collaborative Issue Identification: To ensure coaching is relevant and coachees feel engaged
- Avoidance of Direct Advice: Empowering individuals to develop personalized solutions
- Follow-Up Observations: Opportunistic coaching moments reinforced concepts in real work environments

This approach not only increased engagement but also embedded learning within actual operational contexts, improving transfer and retention.

Typical Coaching Session and Key Topics Covered

PDO's HSE coaching sessions focus on practical, relevant topics designed to enhance safety behaviors and leadership capabilities. Common themes addressed include hazard identification, risk assessment, effective communication and speaking up about hazards, leadership behaviors and safety conversation techniques, incident investigation and near-miss reporting, promoting peer-to-peer safety engagements, fostering psychological safety, and building trust and collaboration between teams and contractors.

A typical coaching session begins with open-ended questions to encourage reflection and discussion of recent site experiences, including incidents or safety observations. Coaches collaborate with participants to identify specific safety challenges and facilitate the development of personalized action plans, avoiding prescriptive advice. Role-playing and practical exercises are often used to build confidence in speaking up and applying risk controls. Sessions conclude with clear commitments and plans for follow-up observations to reinforce learning in daily operations. This hands-on, interactive approach ensures that coaching is embedded in the real work environment, driving meaningful and sustainable safety culture improvements.

Discussion

The integrated HSE coaching, and competency assurance programs implemented by PDO represent a comprehensive, dual-strategy approach to advancing safety culture across both contractor populations and internal leadership. This discussion explores the implications of the findings, situating them within relevant theoretical and practical frameworks.

Integration of Dual Coaching Strategies for Sustainable Safety Culture

PDO's two-pronged coaching approach, targeting Major Contractors, LCCs and SMEs through structured group and one-on-one coaching, alongside internal leadership development via the LPA risk management framework, addresses the complex socio-technical nature of HSE performance. The results demonstrate the importance of harmonizing external contractor capability building with internal leadership competency assurance to promote aligned safety values and behaviors throughout the operational ecosystem.

The high participation rates and positive feedback underscore the effectiveness of tailored coaching interventions to overcome common barriers such as mistrust, fragmented knowledge, and hierarchical distance prevalent in contractor environments (Nielsen, 2014; Provan et al., 2017). This alignment helps bridge the "safety culture gap" often found between operators and contractors in complex industrial settings.

Building Resilient Safety Cultures Through Capability Development

Focusing on capability rather than mere compliance fosters resilience by equipping individuals not only with technical safety knowledge but also with the interpersonal and psychological skills necessary for proactive and adaptive behaviors (Dekker, 2014; Borys, 2019). The "Know, Say, Do" model encapsulates this holistic

capability development, reinforcing the theoretical shift from compliance-based safety management toward a capability and human factors-driven paradigm (Reason, 1997).

The coaching methodology's emphasis on influencing without formal authority and facilitating participative dialogue aligns well with contemporary leadership and safety culture theories that highlight the value of adaptive leadership and collaborative safety management (Weick & Sutcliffe, 2015; Hopkins, 2013). This approach helps embed safety as a shared responsibility rather than a top-down mandate.

Addressing Bureaucracy and Enhancing Influence of HSE Professionals

Consistent with Provan et al. (2017), HSE professionals often face constraints due to bureaucratic workload, which can limit their ability to effect behavioral change. PDO's program intentionally reallocates effort from compliance checklists toward leadership coaching and behavior change facilitation. This transition empowers HSE staff to become trusted advisors and change agents rather than system managers, crucial for cultivating an effective and positive safety culture (Guldenmund, 2000).

Contractor Engagement as a Strategic Lever for System-Wide Improvement

Engaging local contractors and SMEs through coaching represents a strategic investment in the broader safety ecosystem. Contractors traditionally demonstrate varied safety maturity and resource constraints. The coaching program's ability to align contractor safety practices and values with PDO's Goal Zero objectives addresses a critical vulnerability in the supply chain safety landscape.

The documented improvements in near-miss reporting and incident reduction among coached contractors indicate growing ownership and accountability, reinforcing findings from similar case studies in oil and gas sectors where targeted coaching led to measurable safety culture improvements (Mearns et al., 2013; Zohar, 2010).

Challenges

Despite these encouraging outcomes, several challenges were identified:

- **Measuring Long-Term Safety Outcomes:** Safety culture transformation is a long-term endeavor with multifactorial influences. Attribution of incident rate improvements directly to coaching requires extended longitudinal data and consideration of other simultaneous interventions and operational changes
- **Heterogeneity of Contractor Organizations:** The diversity in contractor size, operational scope, and workforce skill levels presents challenges for standardizing coaching content and delivery. Tailoring interventions while maintaining program consistency demands ongoing adaptation and resource allocation
- **Data Quality and Consistency:** Digitization has improved coaching data capture and analysis, yet data integrity depends heavily on coach engagement, accurate reporting, and participant honesty. Variability in data input quality can affect the reliability of program impact assessments
- **Sustaining Engagement:** Maintaining high levels of participation and motivation over time, especially among contractors with limited resources, requires continuous innovation in delivery methods and incentives

Limitations

Several limitations temper the interpretation of findings:

- **Self-Report Bias:** Participant feedback surveys are susceptible to social desirability bias, potentially inflating positive responses about coaching effectiveness and behavioral change.

- **Short-Term Data Window:** The current evaluation covers a relatively brief period (January 2024 to June 2025). Longer-term studies are necessary to validate sustained behavioral and safety outcome improvements
- **Lack of Control Groups:** The absence of control or comparison groups limits the ability to isolate the effects of the coaching intervention from other safety initiatives or external factors
- **Variability in Coaching Delivery:** Differences in coach experience, style, and local contextualization may introduce variability in the coaching quality and impact, complicating standardized evaluation

Summary

Overall, PDO's integrated coaching and competency assurance model exemplifies a forward-thinking, evidence-informed approach to embedding a resilient safety culture across a complex operational landscape. By combining tailored contractor engagement with robust internal leadership development, supported by digital infrastructure, PDO advances safety capability systematically and sustainably.

Conclusions

The integrated HSE coaching and competency assurance program implemented by PDO offers a pioneering model for safety culture transformation that effectively spans both internal leadership and the external contractor ecosystem. The key conclusions from this study are:

1. **Dual-Strategy Effectiveness:** The combination of focused coaching for major Contractors, LCCs and SMEs alongside a structured competency assurance framework LPA for PDO leadership has demonstrated significant improvements in safety climate, leadership behaviors, and workforce capability across diverse operational groups. This dual approach helps harmonize safety values and practices throughout the operational ecosystem, fostering collective ownership and proactive risk management
2. **Capability as a Core Concept:** Emphasizing capability development through the holistic "Know, Say, Do" model aligns technical knowledge with behavioral and interpersonal competencies. This comprehensive focus addresses human factors, psychological safety, and communication skills essential for sustaining positive safety behaviors and adaptive leadership
3. **Digitization as an Enabler:** The strategic use of digital platforms to manage coaching database, competency assessments, and data analytics significantly enhances the program's scalability, consistency, and impact measurement. Digitization facilitates real-time feedback loops and data-driven decision-making, empowering coaches and leadership to continuously refine and optimize safety interventions
4. **Sustainable Safety Culture Development:** Embedding coaching within routine operational practices promotes ongoing learning, trust-building, and shared responsibility for safety. This integration is critical for sustaining culture change and achieving PDO's Goal Zero ambition of zero harm across its operations and contractor networks
5. **Scalability and Replicability:** PDO's model offers a scalable and replicable blueprint for other oil and gas operators or complex industrial enterprises seeking to elevate safety culture through integrated coaching and competency assurance strategies. The program's adaptability to diverse contractor maturity levels and internal leadership needs underscores its broad applicability

This comprehensive study detailed the design, implementation, and evaluation of PDO's HSE coaching program and leadership competency assurance framework. Through over 3,000 coaching sessions engaging 40,000 participants, PDO has fostered measurable improvements in safety knowledge, behaviors, and leadership competencies. Qualitative and quantitative data reveal enhanced safety ownership among

contractors, strengthened leadership influence, and early indicators of improved safety performance, including reduced incident rates and increased near-miss reporting.

The coaching methodology, grounded in behavioral safety, adult learning, and safety culture theories, promotes reflective learning and capability building rather than compliance-focused training. The digital tools supporting the program have been critical in managing scale, ensuring data integrity, and enabling insightful analytics.

While challenges remain, such as measuring long-term impact and managing contractor heterogeneity, the program's success to date affirms the value of integrating coaching and competency assurance within a digitally enabled framework to drive culture change and risk reduction.

Future Directions

Building on the successes and insights from the current program, PDO is committed to evolving its HSE coaching and competency assurance initiatives through several future-focused strategies:

- **Advanced AI and Predictive Analytics Integration:** PDO plans to harness artificial intelligence to analyze coaching session transcripts, assess coaching effectiveness, and identify emerging safety risks proactively. Predictive analytics will enable targeted interventions at sites or teams exhibiting early signs of safety performance decline, fostering anticipatory risk management
- **Expanded Contractor Inclusion:** Extending the coaching program to encompass a broader range of contractors, including subcontractors and informal workforce segments, will further embed safety culture throughout the supply chain. Customized content and delivery methods will be developed to address varying capability levels and operational contexts
- **Longitudinal Impact Studies:** Implementing longer-term evaluations to assess sustained behavioral change and safety outcomes will strengthen the evidence base. Incorporating control groups and benchmarking against industry standards will enhance causal attribution and program optimization
- **Coach Development and Quality Assurance:** Investing in continuous professional development for coaches, including advanced coaching skills from Association of Coaching or the International Coaching Federation, cultural competence, and data literacy, will ensure consistent coaching quality. AI-assisted quality control mechanisms will support ongoing performance monitoring and improvement
- **Cross-Industry Knowledge Sharing:** PDO intends to collaborate with other oil and gas operators, industry bodies, and academic institutions to disseminate learnings, share best practices, and collectively advance HSE coaching methodologies globally

Remarks

PDO's integrated HSE coaching and competency assurance program embodies a transformational journey that bridges leadership and frontline behaviors, technical knowledge and interpersonal skills, compliance and capability. Through deliberate design, rigorous evaluation, and digital empowerment, the program sets a new standard for fostering resilient safety cultures in complex industrial environments. The continued evolution of this approach promises not only enhanced safety performance but also strengthened organizational resilience, workforce engagement, and sustainable operational excellence.

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